SIBYL: Speculative Inference Balancing for XPYD Scheduling in Disaggregated LLM Serving

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Abstract

Disaggregated serving is widely adopted for large language model inference, where requests are assigned to decode instances at prefill start to enable overlapping of KV-cache transmission with computation. The assignment typically aims to maximize performance based on the current loads of decode instances. However, when a request finishes prefilling, the actual load on its assigned decode instance usually differs from that at the time of decision-making. This load discrepancy *makes the assignment suboptimal and often leads to elevated tail latency that violates service-level objectives. The discrepancy stems from two sources: (1) backlog requests may finish decoding before the new request arrives at the assigned instance; and (2) requests currently undergoing prefilling may transition to decoding at any moment. To address this issue, we propose SIBYL, a speculative scheduler that projects per-instance load at the handoff moment and assigns each request to the instance with the least projected load. The projection is driven by a probabilistic workload model that jointly captures the residual lifetime of active tasks and the shadow interference introduced by in-flight prefills. We implement SIBYL as a lightweight, engine-agnostic proxy that requires no modification to existing inference engines. Experiments on real clusters and large-scale simulations show that SIBYL reduces decoding tail latency by up to* 56.9% *and lifts the optimal-assignment ratio from* 43.2% *to* 89.8% *over state-of-the-art approaches.*

1 Introduction

The rapid adoption of large language models (LLMs) [4, 12, 28, 29, 34, 48, 49, 59, 64] has driven the need for high-throughput, low-latency inference systems. LLM inference proceeds in two stages: (1) *prefill*, which processes the full prompt to initialize the key-value cache (KV cache), and (2) *decode*, which autoregressively generates tokens while reading the KV cache. To improve resource utilization and scalability, many modern serving frameworks [7, 20, 31, 35, 36, 39, 65] adopt *disaggregated serving*, which separates prefill and decode into two instance[1] pools with X prefill instances and Y decode instances (XPYD): prefill instances process prompts and produce KV caches, while decode instances consume these caches to generate tokens. The separation of prefill and decode eliminates the mutual interference in mixed workloads, where bursty prefills and concurrent decodes mutually stall each other [1]. This disaggregated design is critical for production-scale serving to meet service level objectives (SLOs) for both time-to-first-token (TTFT) and time-per-output-token (TPOT).

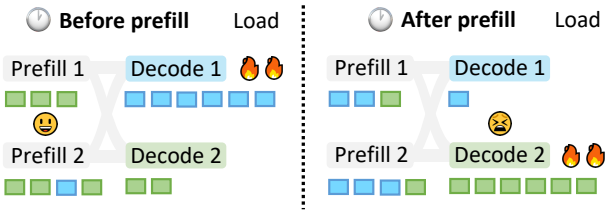

Figure 1. The load discrepancy in disaggregated serving. *Each colored block denotes a request, and the color indicates the decode instance it is bound to before prefill. An assignment at prefill start (Left) becomes suboptimal at decode start (Right) as the cluster state evolves during prefill, causing load imbalance.*

Despite the architectural benefits, disaggregated clusters frequently suffer from severe tail latency, causing SLO violations [53, 62]. This tail-latency issue stems from load imbalance: some instances become overloaded while others remain underutilized. One important source of this imbalance is a fundamental *load discrepancy* introduced by early assignment, which aims to overlap KV cache transmission with computation [35, 36]. This means the scheduler must choose a target decode instance at prefill start, while decoding only begins at the handoff moment (when prefill completes), risking placement on an instance that becomes overloaded during the prefill interval. As illustrated in Figure 1, a scheduler relying on the instantaneous snapshot at prefill start (Left) routes the request to the seemingly idle Decode 2. By the time decoding begins (Right), however, the load distribution has changed. Concurrent arrivals have increased the load on Decode 2, while requests on Decode 1 have completed. As a result, the original assignment is no longer load-balanced and becomes suboptimal.

Existing schedulers fail to mitigate this discrepancy because they rely on instantaneous metrics that become outdated during the prefill interval [13, 20, 24, 31, 39]. Addressing this problem requires shifting from reactive placement

[1]Throughout this paper, an instance hosts a full model replica.

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to predictive look-ahead. However, modeling the exact cluster state at the future handoff moment is non-trivial due to two distinct sources of uncertainty: (1) *Residual Uncertainty*, arising from the remaining lifetime of currently active tasks, which is highly stochastic, especially with the variance in generation lengths; and (2) *Shadow Interference*, arising from requests currently undergoing prefill, which constitute a latent workload that will contend for decode resources at the future handoff moment. Navigating these uncertainties to enable precise early assignment remains an unresolved challenge.

To mitigate the load discrepancy, we introduce Sibyl, a speculative scheduler for disaggregated serving. Instead of relying on instantaneous snapshots, Sibyl maintains a fine-grained *probabilistic workload model* that tracks every active and pending request and forecasts how each decode instance's occupancy evolves over time. The model decomposes the projected load at the future handoff moment τ into three categories: (i) *active tasks* currently decoding, whose residual lifetime at τ is captured via survival analysis; (ii) *pre-handoff shadow tasks* that are still in prefill but are projected to enter decoding before the new request; and (iii) *post-handoff shadow tasks*, prefilling requests whose decoding will overlap with the new request shortly after handoff. For each task, the projected load is expressed as the product of a *physical pressure* term (its KV-cache footprint at τ) and a *survival probability* (the conditional probability of remaining active at τ), estimated online via a discretized survival function that adapts to workload drift without retraining. On each request arrival, Sibyl estimates the prefill duration, queries this model to project per-instance loads at the predicted handoff, and binds the request to the instance with the minimum projected load, thereby preserving the KV-cache transfer-overlap benefits of early assignment while avoiding the load discrepancy pitfalls. We realize Sibyl as a lightweight, framework-agnostic proxy that integrates with mainstream inference engines such as vLLM and SGLang [20, 39] without modifying the internal executors, making speculative scheduling readily deployable in black-box production environments. Across real-cluster experiments and large-scale simulations, Sibyl cuts TPOT tail latency by up to 56.9% over state-of-the-art baselines and lifts the optimal-assignment ratio from 43.2% to 89.8% on heavy-tailed workloads.

We make the following contributions:

- We identify a *load discrepancy* between scheduling (prefill start) and execution (decode start) as a key contributor to decode-side tail latency in disaggregated serving. We further attribute this discrepancy to two sources: residual uncertainty of active tasks and shadow interference from in-flight prefills.
- We develop a probabilistic workload model that decomposes projected decode load into active, pre-handoff shadow, and post-handoff shadow components. Each component is modeled as a physical-pressure term weighted by an online-estimated survival probability, enabling robust scheduling under the heavy-tailed uncertainty of LLM generation lengths.
- We implement Sibyl as a lightweight, framework-agnostic scheduler proxy that integrates with mainstream inference engines without invasive modifications, making speculative early assignment practical for black-box production deployments.
- We evaluate Sibyl through real-cluster experiments and large-scale simulations with up to 128 decode instances. Sibyl reduces P99/P99.9 TPOT tail latency by up to 56.9% over state-of-the-art baselines and increases the optimal-assignment ratio from 43.2% to 89.8% on heavy-tailed workloads.

2 Background and Motivation

2.1 The Heterogeneity of Disaggregated Serving

LLM inference proceeds in two phases. Given an input prompt of n tokens $x_{1:n} = (x_1, \ldots, x_n)$, the *prefill* phase processes all tokens in parallel and produces the per-layer key/value tensors collectively known as the KV cache:

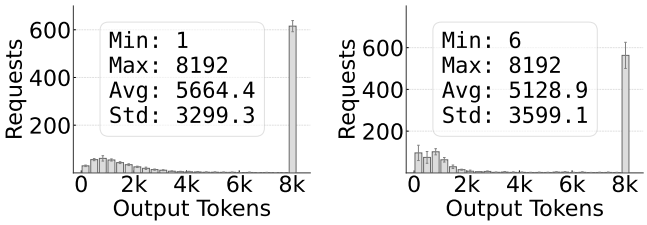
$$\mathrm{KV}_{1:n} = f_{\mathrm{prefill}}(x_{1:n}). \tag{1}$$

Per-layer computation is dominated by dense matrix multiplications over all n tokens at once, so prefill is *compute-bound* and bursty. The *decode* phase then autoregressively emits output tokens one at a time. At step t, given the existing KV cache, the model produces the next token

$$x_{n+t} = f_{\mathrm{decode}}\big(x_{n+t-1}, \mathrm{KV}_{1:n+t-1}\big), \tag{2}$$

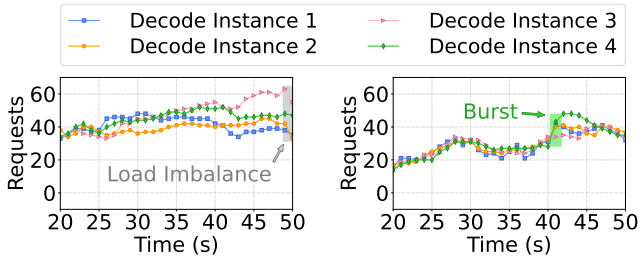
and the newly produced K, V entries are appended to the cache. Per-step computation is small and roughly constant, whereas memory traffic for reading the KV cache grows linearly with the running context length $n + t$, making decode *memory-bandwidth bound* and latency-sensitive [16, 17]. These contrasting hardware profiles are reflected in the two SLO metrics that govern serving quality: TTFT, determined by prefill latency, and TPOT, determined by per-step decode latency.

To exploit this heterogeneity, modern serving systems [18, 27, 31, 35, 36, 44, 65] adopt *disaggregated serving*, separating the two phases into dedicated instance pools backed by heterogeneous hardware tailored to each phase's characteristics. We consider an *XPYD* cluster comprising X prefill instances $\{P_1, \ldots, P_X\}$ and a set $\mathcal{D} = \{D_1, \ldots, D_Y\}$ of Y decode instances, each hosting a full model replica. Upon arrival, a request k is first scheduled to a prefill–decode pair (P_i, D_j), executes its prefill on P_i, has its KV cache streamed to D_j in parallel with the prefill computation, and finally generates output tokens on D_j [35, 36]. Crucially, the assignment (P_i, D_j) is committed at scheduling time, before prefill starts, so that KV cache transmission can be overlapped with prefill



(a) ShareGPT **(b)** Zeta

Figure 2. Output lengths on different datasets using Qwen3-32B [59]. *The workloads are evaluated with reasoning enabled and a maximum output limit of 8192 tokens.*



(a) RR Running Requests **(b)** ILS Running Requests

Figure 3. Limitations of baseline scheduling policies under a ShareGPT workload. *(a) RR fails to account for request heterogeneity, leading to significant load imbalance. (b) ILS ignores the prefill-decode time gap, blindly routing multiple requests to the same seemingly idle instance, which causes transient load spikes.*

computation rather than serialized after it. This architecture confines compute-heavy prefill bursts to the prefill pool and shields the latency-sensitive decode pool $\mathcal{D}$ from compute interference, while letting X and Y scale independently along their respective SLOs. These advantages have driven its widespread adoption in production environments [7, 36, 57].

2.2 The Load Discrepancy in Disaggregated Serving

Disaggregated serving introduces a *load discrepancy* between scheduling and execution. To make KV-cache transmission in parallel with computation, requests must be assigned to a decode instance before prefill starts. The load of decode instances, however, can change drastically after the prefill period, due to the influence of "uncertain" evolving loads. We identify two sources of this uncertainty:

(1) Residual Uncertainty from Active Tasks. At the moment of scheduling, instances host ongoing decode tasks with uncertain remaining lifetimes. This is further exacerbated by the emergence of reasoning-intensive models (*e.g.,* Chain-of-Thought [33, 54]), where generation length is dictated by the complexity of the logic chain rather than simple conversational patterns. We validate this using Qwen3-32B [59], as shown in Figure 2. The workloads exhibit extreme variance driven by reasoning steps: ShareGPT [41] displays a massive deviation ($\sigma \approx 3317$) with an average output length of 5657 tokens, frequently saturating the 8192 token limit. The Zeta dataset [60] exhibits a similarly large standard deviation ($\sigma \approx 3492$). As a result, residual decode load is inherently difficult to predict, often leading to severe errors when estimating future load.

(2) Interference from Shadow Tasks. Equally critical is the *shadow workload*, which refers to requests assigned to a decode instance but still currently undergoing prefill. Although not yet decoding, these tasks impose two distinct risks to the given new request upon decoding: tasks transitioning to the decode phase *prior* to the new request (impose immediate contention) and those joining the decode phase during mid-execution (introduce delayed interference). Failing to account for these risks blinds the scheduler to impending pressures, causing concurrent requests to converge on the same seemingly idle instance.

	RPS				
Metric	2	4	6	8	10
Load Shift Rate (%)	8.4	13.0	22.0	28.5	36.1
Bind Error Rate (%)	21.3	29.2	34.5	40.0	42.4

Table 1. ILS suffers from substantial load shift and binding errors as the request rate grows on a ShareGPT workload. *(1) Load shift rate: the average relative change in load on the ILS-bound instance between prefill start and handoff. (2) Bind error rate: the fraction of requests whose bound instance is no longer the least-loaded one at handoff.*

2.3 Limitations of Existing Solutions

Most existing schedulers are myopic, relying on static rules or instantaneous load snapshots, and thus become unreliable under stochastic, fast-evolving workloads (Figure 3). Migration can partially correct imbalance, but it is reactive and introduces non-trivial transfer overhead. We next summarize representative strategies and their limitations.

(1) Round-Robin (RR). Being load-agnostic, RR [20] fails to accommodate the heavy-tailed heterogeneity of LLM generation. As shown in Figure 3(a), the stochastic arrival of long-decode tasks can lead to disproportionate backlog accumulation on specific instances. Since RR continues to blindly route requests to these instances, it exacerbates load imbalance and leads to severe tail latency.

(2) Instantaneous Load Scheduling (ILS). Strategies such as least outstanding requests [24, 31] route new tasks to the instance with the minimum instantaneous load. Because the snapshot used at prefill start becomes stale during the prefill interval, ILS suffers from two parallel failure modes that are both manifestations of this myopia. *First, each individual binding decision is unreliable*: the bound instance is

Model	Attn.	CW	KV/tok	KV@CW	Mig.
Qwen3-32B [59]	GQA†	128K	256 KB	32.0 GB	172 ms
Qwen3-235B-A22B [59]	GQA	128K	188 KB	23.5 GB	126 ms
Hy3-preview (300B) [51]	GQA	256K	320 KB	80.0 GB	400 ms
DeepSeek-V3 (671B) [7]	MLA‡	128K	34.3 KB	4.30 GB	23 ms
DeepSeek-V4 (1.6T) [6]	CSA§+HCA¶	1M	4.8 KB	4.81 GB	24 ms

†Grouped-Query Attention; ‡Multi-head Latent Attention; §Compressed Sparse Attention; ¶Heavily Compressed Attention.

Table 2. KV-cache volume and migration time for a single in-flight request at the model's maximum context window (CW). *KV-cache stored at native precision (BF16 for Qwen3 and Hy3, FP8 for DeepSeek).*

frequently no longer the least-loaded one by handoff. We quantify this on a ShareGPT trace using two metrics (Table 1): the *load shift rate* measures the average relative change in load on the ILS-bound instance between prefill start and handoff, while the *bind error rate* measures the fraction of requests whose bound instance is no longer the least-loaded one at handoff. Both rise sharply with the request rate: at 10 RPS, the load on the bound instance shifts by 36% on average, and ILS picks the wrong instance for 42% of requests. *Second, concurrent decisions become correlated under bursty traffic*: because all in-flight requests are routed against the same stale snapshot, they converge onto the same seemingly idle instance, producing the sharp congestion spikes visible in Figure 3(b).

(3) Rescheduling and Migration. While migration mechanisms [47, 53] can mitigate imbalance, they are inherently reactive and costly. Migrating an in-flight request requires transferring its *entire* KV cache across nodes, whose volume grows linearly with the running context length. Table 2 quantifies this for representative models at their maximum context window: a single long-context request can require migrating tens of GBs of KV state, reaching 32 GB for Qwen3-32B and 4.8 GB for DeepSeek-V4, while even aggressively compressed MLA models still require transferring several GBs. Even over a fast 200 GB/s inter-node link (4× ConnectX-7, a common decode-side configuration on H20-based servers), migration latency can exceed 100 ms for large-context requests, reaching up to 400 ms in our examples. Worse, these times are highly optimistic upper bounds: in practice the migration traffic contends fiercely with the incoming KV-cache traffic from prefill instances and with the expert-parallel (EP) all-to-all traffic inside the decode instance, so the effective stall is substantially larger. Yet even this best case already dwarfs the per-token TPOT SLO (typically ~50 ms), stalling the very request it aims to rescue. Beyond this bandwidth penalty, such mechanisms also mandate invasive modifications to engine internals, rendering them impractical for standard black-box deployments.

Overall, existing designs are either myopic or migration-heavy, motivating Sibyl as a proactive, lightweight method to mitigate load discrepancy in disaggregated serving.

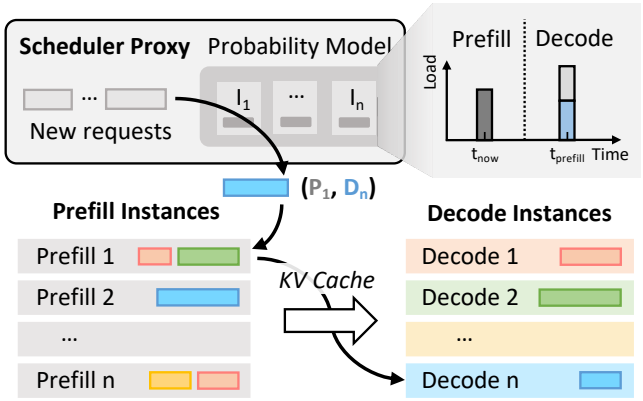


Figure 4. System overview of Sibyl. *The probability-driven global scheduler proxy can predict future decode loads. It assigns new requests to prefill (P_i) and decode (D_j) instances based on projected availability at the estimated handoff time.*

3 Design Overview

3.1 System Architecture

As shown in Figure 4, Sibyl interposes a thin *global scheduler proxy* on the control path of an existing XPYD cluster. For *transparency*, it separates control from data: prefill and decode instances run *unmodified* engines (*e.g.*, vLLM, SGLang [20, 39]) and stream KV caches peer-to-peer as before, while the proxy only observes lightweight metadata (prompt lengths and decode start/finish events) and emits one placement per request. Sibyl thus drops in as a black-box replacement for the cluster's stock load balancer, with no executor changes.

On each arrival, the proxy chains three steps. *(i)* It estimates the request's prefill duration, which is near-determ inistic given the prompt length [26, 57, 65], to pin down the future *handoff moment* τ at which the request will actually begin decoding. *(ii)* It consults a *probabilistic workload model* that, for every decode instance, predicts how busy that instance will be *at* τ, accounting both for requests still running then and for requests currently in prefill that will have joined decoding by then. *(iii)* It binds the request to the prefill–decode pair (P_i, D_j) whose decode instance is predicted to be *least loaded* at τ, and marks it as pending there so that simultaneous decisions stay consistent. Prefill then runs on P_i with its KV cache streamed to D_j in the background, and decoding starts at handoff. By deciding at prefill start yet optimizing for the predicted state at τ, Sibyl keeps the transfer-overlap benefit of early assignment while still balancing load at the moment decoding actually begins. This design, however, hinges on a single hard question: how to faithfully project the decode load at τ across the gap between when the decision is made (prefill start) and when it takes effect (handoff), an interval over which the cluster state can shift substantially. We next present the principle

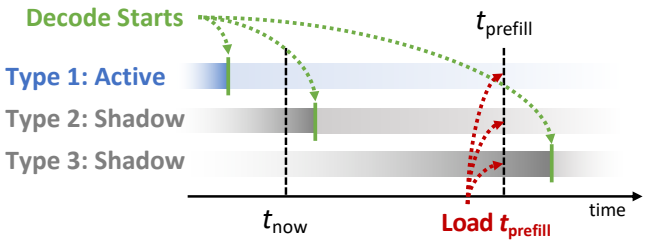


Figure 5. Speculative load projection. *The scheduler estimates instance occupancy at τ (the future handoff moment, also referred to as $t_{prefill}$), when the new task arrive at t_{now}, by aggregating the residual active load and the shadow load from pending tasks.*

by which Sibyl bridges this gap (§3.2), and formalize the resulting workload model (§4).

3.2 Design Philosophy: "Bridging" the Temporal Gap

To mitigate the load discrepancy, Sibyl optimizes for the projected cluster state at the future handoff moment τ rather than the current snapshot t_{now}. As shown in Figure 5, the scheduler aggregates potential load (introduced by three types of tasks) to identify the instance with the minimal projected load at the handoff moment τ:

- **Type 1: Active Tasks.** Requests currently being decoded. Their residual load at τ is stochastic, derived from the survival probability during prefill.
- **Type 2: Pre-Handoff Shadow.** Concurrent prefill tasks projected to start decoding *before* τ. These form hidden contention that is invisible at t_{now} but will be occupying resources at the handoff.
- **Type 3: Post-Handoff Shadow.** Concurrent prefill tasks projected to start decoding *after* τ. Although arriving later, they pose a potential interference risk that may constrain near-term availability.

By aggregating the projected impact of these workloads, Sibyl constructs a comprehensive view of the *future* load.

4 Probabilistic Workload Modeling

We establish a probabilistic framework to formulate the scheduling decision as a load minimization problem at the handoff moment. For reference, the key notations are summarized in Table 3.

4.1 Objective Function

Let $\mathcal{D}$ denote the set of decode instances. For a candidate instance $j \in \mathcal{D}$, the total projected load $\mathcal{L}_j(\tau)$ at the future handoff time τ is defined as the sum of the residual load from currently running tasks and the projected load from pending tasks (*i.e.,* requests currently undergoing prefill that are already bound to instance j):

$$\mathcal{L}_j(\tau) = \sum_{k \in Q_j^{\text{run}}} \mathcal{L}_{\text{I}}(k,\tau) + \sum_{k \in Q_j^{\text{pend}}} \left(\mathbb{I}_{\tau_k \le \tau} \mathcal{L}_{\text{II}}(k,\tau) + \mathbb{I}_{\tau_k > \tau} \mathcal{L}_{\text{III}}(k,\tau) \right) \tag{3}$$

where Q_j^{run} and Q_j^{pend} denote the sets of active and pending tasks on instance j, respectively. The terms $\mathcal{L}_{\text{I}}$, $\mathcal{L}_{\text{II}}$, and $\mathcal{L}_{\text{III}}$ correspond to the projected loads of active, pre-handoff shadow, and post-handoff shadow tasks, as detailed in §3.2. $\mathbb{I}_{\text{condition}}$ denotes the indicator function, which equals 1 if the condition holds and 0 otherwise. Based on this metric, the scheduler identifies the optimal instance

$$j^* = \arg\min_{j \in \mathcal{D}} \mathcal{L}_j(\tau), \tag{4}$$

thereby directing the request to the instance with the minimum projected load at the handoff moment. The following sections define the individual load terms $\mathcal{L}_{\text{I}}$, $\mathcal{L}_{\text{II}}$, and $\mathcal{L}_{\text{III}}$.

Symbol	Description
System State & Scheduling	
t_{now}	Current scheduling timestamp
τ	Projected handoff moment, $t_{\text{now}} + \delta_{\text{prefill}}$
$\mathcal{D}$, j	Decode instance set; candidate $j \in \mathcal{D}$
$Q_j^{\text{run}}, Q_j^{\text{pend}}$	Active / pending tasks on j
v_k, $\bar{v}_{\text{sys}}$	Per-task / system-average decode rate
Probabilistic Primitives	
L_k, l_k^{in}	Generation-length r.v.; input prompt length
$l_k(t)$	Generated length of task k at time t
$\mathcal{M}_k(\tau)$	Physical pressure: $l_k^{\text{in}} + l_k(\tau)$
$\mathcal{S}(x), \hat{\mathcal{S}}(x)$	Survival function and its online estimator
$\mathcal{P}_k(\tau)$	Conditional survival prob. of k at τ
Speculative Load Composition	
$\mathcal{L}_j(\tau)$	Total projected load on j (to minimize)
$\mathcal{L}_{\text{I,II,III}}(k,\tau)$	Loads from Type 1 / 2 / 3 tasks

Table 3. Key notations used in the probabilistic workload modeling.

4.2 Modeling Primitives

To compute $\mathcal{L}_{\text{I, II}}$ in Eq. (3), we introduce a mechanistic model built upon two primitives: *Physical Pressure* ($\mathcal{M}$) and *Survival Probability* ($\mathcal{P}$).

Primitive 1: Physical Pressure ($\mathcal{M}$). We define physical pressure $\mathcal{M}_k(\tau)$ as the request's total token occupancy to

quantify its hardware cost:

$$\mathcal{M}_k(\tau) = l_k^{\text{in}} + l_k(\tau) \tag{5}$$

where l_k^{in} denotes the prompt length and $l_k(\tau)$ represents the generated length at time τ. Since LLM decoding is memory-bandwidth bound [35, 65], the execution overhead is dominated by loading the KV cache, which scales linearly with the total sequence length. Thus, $\mathcal{M}$ effectively captures the hardware cost.

Primitive 2: Conditional Survival Probability ($\mathcal{P}$). This component quantifies the *likelihood* of a request remaining active at a specific future moment. Predicting exact generation lengths is inherently unstable [53, 55]. Instead, we model task completion uncertainty using the survival function, which directly captures the probability of a request remaining active beyond a given time. Let L_k denote the generation length of request k. We define the survival function as $\mathcal{S}(x) = \mathbb{P}(L_k > x)$ for a length threshold x, and assume that $\{L_k\}$ are i.i.d. samples drawn from a common distribution. For a request k with current generated length $l_k(t_{\text{now}})$, the probability $\mathcal{P}_k(\tau)$ that it survives up to the projected generated length $l_k(\tau)$ is:

$$\begin{aligned}\mathcal{P}_k(\tau) &= \mathbb{P}(L_k > l_k(\tau) \mid L_k > l_k(t_{\text{now}})) \\ &= \frac{\mathbb{P}(L_k > l_k(\tau))}{\mathbb{P}(L_k > l_k(t_{\text{now}}))} = \frac{\mathcal{S}(l_k(\tau))}{\mathcal{S}(l_k(t_{\text{now}}))}\end{aligned} \tag{6}$$

To compute Eq. (6) practically, we approximate the theoretical function $\mathcal{S}$ using an online estimator $\hat{\mathcal{S}}$. Instead of tracking every integer length, we discretize the output length space into coarse-grained intervals (buckets) with a fixed step size Δ. The estimator entries $\hat{\mathcal{S}}(l)$ are maintained only for discrete boundaries $l \in \{\Delta, 2\Delta, \dots\}$. Upon each request completion with final length L_{final}, we update the estimated probabilities for all relevant buckets ($l \le L_{\text{final}}$) using an exponential moving average (EMA):

$$\hat{\mathcal{S}}(l) \leftarrow \alpha \cdot \hat{\mathcal{S}}(l) + (1-\alpha) \cdot \mathbb{I}_{L_{\text{final}} \ge l} \tag{7}$$

where $\alpha \in (0,1)$ is the smoothing factor. This online update prioritizes recent observations, allowing the estimator to refine $\hat{\mathcal{S}}$ to reflect the current output length distribution.

4.3 Speculative Load Composition

Using these primitives, we define the projected load for tasks potentially active at τ (*i.e.,* Type 1 and Type 2) as its expected load at τ, as established in Eq. (8).

$$\mathcal{L}(k,\tau) = \mathcal{M}_k(\tau) \cdot \mathcal{P}_k(\tau), \qquad k \in \text{Type 1/2}. \tag{8}$$

Derivation. We define the system load at time τ as the aggregate decoding load contributed by all active tasks:

$$\mathbb{L}(\tau) = \sum_k \mathcal{M}_k(\tau)\, \mathbb{I}_{(\tau_k \le \tau,\, T_k > \tau)}, \tag{9}$$

where τ_k and T_k denote the decode start time and completion time of task k, respectively. The indicator function ensures that only tasks that have started decoding but have not yet completed contribute to the instantaneous load.

Conditioned on the system state at the current time t_0, taking expectation on both sides yields

$$\mathbb{E}_0[\mathbb{L}(\tau)] = \sum_k \mathbb{E}_0\left[\mathcal{M}_k(\tau)\, \mathbb{I}_{(\tau_k \le \tau,\, T_k > \tau)}\right]. \tag{10}$$

For an active decoding task, the physical pressure evolves linearly over the short prediction interval:

$$\mathcal{M}_k(\tau) = l_{k,0} + v_k(\tau - t_0), \tag{11}$$

where $l_{k,0}$ is the accumulated load at time t_0 and v_k is the observed decoding rate. Since both quantities are deterministic conditioned on the current system state, we can factor them out of the expectation:

$$\mathbb{E}_0[\mathbb{L}(\tau)] = \sum_k \big(l_{k,0} + v_k(\tau - t_0)\big)\, \mathbb{P}_0(T_k > \tau \mid \tau_k \le \tau). \tag{12}$$

Next, observe that a task survives beyond time τ if and only if its total decoding length exceeds the generated length at τ. Therefore,

$$\mathbb{P}_0(T_k > \tau) = \mathbb{P}_0(L_k > l_k(\tau)), \tag{13}$$

where L_k denotes the random total decoding length of task k. Substituting this relation into Eq. (12), we obtain

$$\mathbb{E}_0[\mathbb{L}(\tau)] = \sum_k \underbrace{\big(l_{k,0} + v_k(\tau - t_0)\big)}_{\mathcal{M}_k(\tau)} \cdot \underbrace{\mathbb{P}_0(L_k > l_k(\tau))}_{\mathcal{P}_k(\tau)}, \tag{14}$$

which yields Eq. (8).

For Type 1 and Type 2 tasks, the formulation differs only in the reference state used for projection (*i.e.,* the choice of t_0, $l_{k,0}$, and v_k); the expected-load expression in (14) remains identical. In contrast, Type 3 tasks do not begin decoding until after τ, contributing zero instantaneous load at the handoff. Consequently, instead of applying Eq. (8), we model $\mathcal{L}_{\text{III}}(k,\tau)$ as a lookahead interference penalty to capture near-future contention.

4.3.1 Type 1: Active Tasks ($t_{\text{start}} \le t_{\text{now}}$).

For tasks already in decoding, we leverage their observed execution statistics for projection. We project the token count using the task's specific decoding rate v_k, and compute the conditional survival probability directly from our estimator $\hat{\mathcal{S}}$:

$$\mathcal{L}_{\text{I}}(k,\tau) = \underbrace{(l_k^{\text{in}} + l_k(\tau))}_{\mathcal{M}} \cdot \underbrace{\frac{\hat{\mathcal{S}}(l_k(\tau))}{\hat{\mathcal{S}}(l_k(t_{\text{now}}))}}_{\mathcal{P}} \tag{15}$$

where $l_k(\tau) = l_k(t_{\text{now}}) + v_k \cdot (\tau - t_{\text{now}})$ is obtained by linearly projecting the generated length using the task's observed effective decoding rate v_k[2] over the short interval $[t_{\text{now}}, \tau]$. The probability term $\mathcal{P}$ captures the likelihood that task k remains active from its current generation length $l_k(t_{\text{now}})$ to the projected generation length $l_k(\tau)$.

[2]Calculated as generated length divided by decoding time.

Algorithm 1: SIBYL

Input: Request r, Time t_{now}, Instances $\mathcal{D}$, Estimator $\hat{\mathcal{S}}$

1 **Function** `ScheduleRequest(`r, t`)`**:**
2 $\quad\tau \leftarrow t_{\text{now}} + \text{EstPrefill}(r.length)$;
3 $\quad$**for** *each instance* $j \in \mathcal{D}$ **do**
4 $\quad\quad\mathcal{L}_j, \mathcal{L}_{\text{I}}, \mathcal{L}_{\text{II}}, \mathcal{L}_{\text{III}} \leftarrow 0$;
5 $\quad\quad$**for** $k \in Q_j^{run}$ **do**
6 $\quad\quad\quad l_k(\tau) \leftarrow l_k(t_{\text{now}}) + v_k \cdot (\tau - t_{\text{now}})$;
7 $\quad\quad\quad\mathcal{M} \leftarrow l_k^{\text{in}} + l_k(\tau)$;
8 $\quad\quad\quad\mathcal{P} \leftarrow \hat{\mathcal{S}}(l_k(\tau))/\hat{\mathcal{S}}(l_k(t_{\text{now}}))$;
9 $\quad\quad\quad\mathcal{L}_{\text{I}} \leftarrow \mathcal{L}_{\text{I}} + \mathcal{M} \times \mathcal{P}$;
10 $\quad\quad$**for** $k \in Q_j^{pend}$ **do**
11 $\quad\quad\quad\tau_k \leftarrow k.arrival + \text{EstPrefill}(k.length)$;
12 $\quad\quad\quad\Delta_{\text{gap}} \leftarrow (\tau - \tau_k) \cdot \bar{v}_{sys}$;
13 $\quad\quad\quad$**if** $\Delta_{gap} > 0$ **then**
14 $\quad\quad\quad\quad\mathcal{M} \leftarrow l_k^{\text{in}} + \Delta_{\text{gap}}$;
15 $\quad\quad\quad\quad\mathcal{P} \leftarrow \hat{\mathcal{S}}(\Delta_{\text{gap}})$;
16 $\quad\quad\quad\quad\mathcal{L}_{\text{II}} \leftarrow \mathcal{L}_{\text{II}} + \mathcal{M} \times \mathcal{P}$
17 $\quad\quad\quad$**else**
18 $\quad\quad\quad\quad\mathcal{L}_{\text{III}} \leftarrow \mathcal{L}_{\text{III}} + \text{ReLU}(l_k^{\text{in}} + \Delta_{\text{gap}})$;
19 $\quad\quad\mathcal{L}_j \leftarrow \mathcal{L}_{\text{I}} + \mathcal{L}_{\text{II}} + \mathcal{L}_{\text{III}}$;
20 $\quad$**return** $\arg\min_j \mathcal{L}_j$;

4.3.2 Type 2: Pre-Handoff Shadow Tasks ($\tau_k \le \tau$).

Let τ_k denote the projected start time of task k. We model pending tasks satisfying $\tau_k \le \tau$ as pre-handoff shadow tasks. Since these tasks become active before the new request arrives, they will occupy resources at moment τ. Unlike Type 1 tasks where per-task decoding rate is observable, pending tasks lack runtime history. We rely on the system-wide average speed $\bar{v}_{\text{sys}}$ to estimate the potential resource consumption, and define the projected generated length as $l_k(\tau) = (\tau - \tau_k) \cdot \bar{v}_{\text{sys}}$. For Type 2 tasks, the probability term is computed from the start state at τ_k, where $l_k(\tau_k) = 0$. Thus, the conditional survival probability up to $l_k(\tau)$ reduces to $\hat{\mathcal{S}}(l_k(\tau))/\hat{\mathcal{S}}(0)$, which simplifies to $\hat{\mathcal{S}}(l_k(\tau))$ since $\hat{\mathcal{S}}(0) = 1$. Thus, the expected load for Type 2 tasks is:

$$\mathcal{L}_{\text{II}}(k, \tau) = \underbrace{(l_k^{\text{in}} + l_k(\tau))}_{\mathcal{M}} \cdot \underbrace{\hat{\mathcal{S}}(l_k(\tau))}_{\mathcal{P}} \tag{16}$$

4.3.3 Type 3: Post-Handoff Shadow Tasks ($\tau_k > \tau$).

For tasks projected to start after the handoff, we use an *interference-decay* model to capture their near-future contention with the post-handoff execution window. Although these tasks are inactive at the handoff moment, they will start decoding shortly after and may overlap with the decode of the current request, thereby introducing additional interference. We model this interference by treating the prompt length l_k^{in} as the base interference. We then linearly discount this based on the temporal distance $(\tau_k - \tau)$, using the system-average decode rate $\bar{v}_{\text{sys}}$. Intuitively, tasks starting soon after τ have limited temporal slack and are more likely to overlap with ongoing requests, whereas later arrivals experience weaker contention. We therefore clamp the discounted value at zero to ignore sufficiently distant arrivals:

$$\mathcal{L}_{\text{III}}(k, \tau) = \text{ReLU}\big(l_k^{\text{in}} - \bar{v}_{\text{sys}}\,(\tau_k - \tau)\big). \tag{17}$$

4.4 Scheduling Algorithm

Algorithm 1 outlines the scheduling procedure and runs in $O(|\mathcal{D}| + N_{\text{total}})$ time. Here, $|\mathcal{D}|$ is the number of candidate decode instances, and N_{total} is the number of requests traversed for load aggregation (active decodes and pending shadows). The procedure begins by establishing the target handoff time τ using the new request's estimated prefill latency (Line 2). For each candidate instance, the algorithm then aggregates the projected load from two distinct sources. First, it projects the state of currently active tasks (Lines 6–9), weighting their estimated physical pressure by the corresponding conditional survival probability; Second, it evaluates pending shadow tasks (Lines 11–18) by estimating their start times and separately handling: Type 2 expected loads ($\tau_k \le \tau$) and Type 3 future-load penalties ($\tau_k > \tau$). Finally, the algorithm selects the instance that minimizes the resulting cumulative congestion (Line 20).

5 Implementation

SIBYL is implemented as a lightweight, non-intrusive middleware that operates entirely in the control plane. This framework-agnostic design allows SIBYL to be deployed alongside mainstream inference engines, such as vLLM and SGLang [20, 39], without requiring invasive modifications to the underlying kernels or model executors.

Standalone Architecture. Similar to the reference implementation in vLLM's disaggregated serving example,[3] SIBYL functions as a standalone HTTP proxy. This decoupled architecture allows SIBYL to manage request routing and scheduling decisions centrally while maintaining compatibility with standard OpenAI-compatible API interfaces.

State Synchronization. To perform speculative scheduling, SIBYL maintains a real-time view of the cluster state. Instead of relying on complex protocols, SIBYL utilizes the native monitoring interfaces provided by vLLM. Specifically, the proxy periodically polls the `/metrics` endpoint of each decode instance via HTTP `GET` requests. By parsing these metrics, which include the number of running requests, KV cache usage, and iteration stats, SIBYL updates its internal probabilistic workload model with negligible overhead.

[3] Specifically, `disagg_proxy_p2p_nccl_xpyd.py`.

Prefill Time Estimation. Projecting the handoff moment τ hinges on estimating each request's prefill completion time, a primitive shown to be practical at production scale (*e.g.,* Aegaeon [57] on Alibaba Cloud). Sibyl exploits the regular structure of production XPYD deployments, where prefill instances run with `batch_size=1` and chunked prefill, so every step processes a fixed `chunk_size` tokens; our experiments adopt the same configuration. Under non-prefill-dominated workloads, the per-step latency varies little, making prefill timing reliable to predict. A request of L prompt tokens thus takes $\lceil L/\texttt{chunk\_size} \rceil$ steps, completing after its own steps plus those queued ahead on the same instance. We profile the per-step latency offline and fit a low-degree polynomial ($R^2 = 0.998$; Appendix C.4).

Enabling Reasoning in Benchmarks. Standard benchmarking tools (*e.g.,* `vllm bench`) often lack native support for passing engine-specific parameters required by reasoning models. To evaluate reasoning capabilities (*e.g.,* Chain-of-Thought) systematically, we implemented a request injection mechanism within the Sibyl proxy. For every incoming inference request, the proxy intercepts the JSON payload and automatically injects the necessary configuration to trigger the model's reasoning mode. Specifically, we modify the request body to include:

```
request_data["extra_body"]["chat_template_kwargs"]=
    {"enable_thinking": True}
```

This ensures that all requests processed during our evaluation, regardless of the client-side configuration, correctly trigger the reasoning logic in the reasoning model.

6 Evaluation

6.1 Experimental Setup

We first summarize the experimental settings in this section. Full details are provided in Appendix C.

Testbed. We conduct our evaluation on an NVIDIA H800-based cluster. For all datasets, we deploy a 4P4D disaggregated configuration, with four prefill instances and four decode instances. In all scenarios, each GPU hosts a replica of a single model. For the interconnect, we leverage vLLM's native `P2pNcclConnector` [32] to enable efficient layer-wise KV cache transmission between prefill and decode instances.

Models. We evaluate Sibyl using Qwen3-8B [59], a widely adopted open-weights model in production environments. This model represents a typical single-GPU workload that fits comfortably within the H800's on-device memory (80 GB), eliminating the communication overhead of tensor parallelism [43]. To assess generality across model scales and families, we further evaluate Sibyl on three additional open-weights models: Qwen3-32B [59], Qwen2.5-7B-Instruct [50], and Llama-3-8B-Instruct [2], with full results deferred to Appendix D.

Datasets. We evaluate Sibyl across four distinct datasets to cover a range of inference scenarios:

- `Random`, a synthetic workload generated by the built-in `vLLM bench` benchmark with uniformly distributed input and output lengths, used for controlled micro-benchmarking of system performance.
- ShareGPT [41], a de facto standard dataset comprising real-world user conversations, exhibiting the heavy-tailed length distribution typical of chat services.
- Zeta [60], a dataset of predictive code-editing tasks from Zed Industries, used to assess performance on specialized programming assistant workloads.
- NuminaMath [21], a large-scale collection of competition-style mathematics problems paired with chain-of-thought solutions, which stresses Sibyl under reasoning-intensive workloads with long and highly variable output lengths.

Together, these datasets cover a broad spectrum of mainstream LLM serving scenarios, including synthetic micro-benchmarking, conversational chat, programming assistance, and mathematical reasoning.

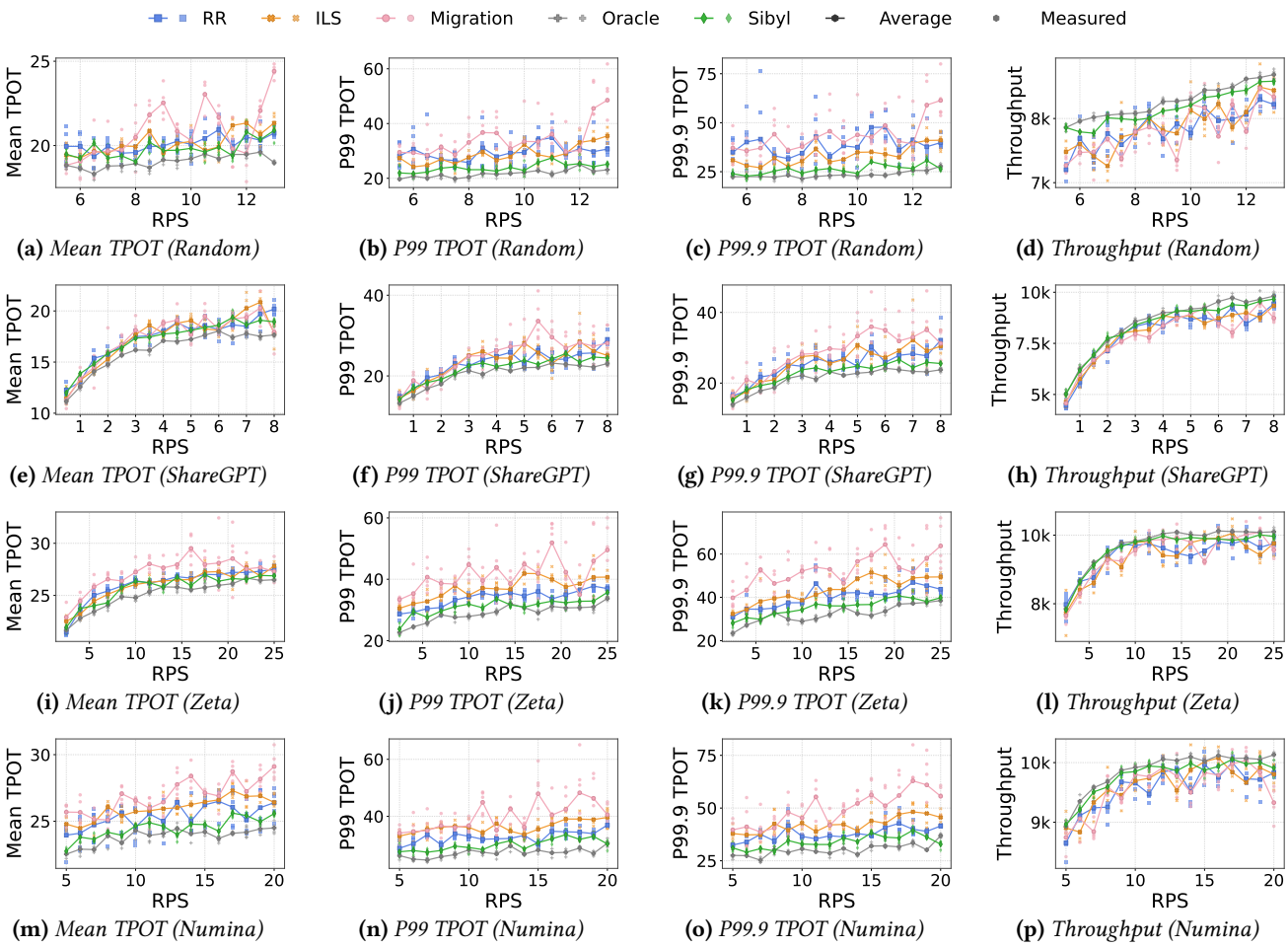
Baselines. We compare Sibyl against four representative policies that span the design spectrum from load-agnostic dispatch to oracle-guided scheduling:

- *Round-Robin (RR)* [20], a static, load-agnostic approach that dispatches requests cyclically, which is also the default scheduling policy adopted in serving frameworks such as vLLM and SGLang.
- *Instantaneous Load Scheduling (ILS)* [24, 31], a greedy policy that routes each request to the instance with the fewest active requests.
- *Migration* [47, 53], a reactive rebalancing baseline that adapts the runtime migration mechanism of Llumnix [47] to the disaggregated setting, transferring KV caches from overloaded to under-utilized decode instances after early assignment.
- *Oracle*, an idealized variant of Sibyl that replaces the survival-based residual-lifetime estimator with the ground-truth remaining decode length of every active task, serving as the practical performance upper bound of our framework.

6.2 Overall System Performance

Figure 6 reports the end-to-end performance of Sibyl against the three deployable baselines (RR, ILS, and Migration) on four workloads, together with the idealized Oracle variant as an upper bound. For every metric, we report the *average / maximum* percentage improvement of Sibyl (or Oracle) over the corresponding baseline across all tested request rates.

Tail latency. Sibyl's gains are most pronounced at the tail, where load-discrepancy-induced hotspots translate directly into SLO violations. On the bursty Random workload, where concurrent prefills repeatedly target the same seemingly idle decode instance, Sibyl reduces P99 / P99.9



(a) *Mean TPOT (Random)* **(b)** *P99 TPOT (Random)* **(c)** *P99.9 TPOT (Random)* **(d)** *Throughput (Random)*

(e) *Mean TPOT (ShareGPT)* **(f)** *P99 TPOT (ShareGPT)* **(g)** *P99.9 TPOT (ShareGPT)* **(h)** *Throughput (ShareGPT)*

(i) *Mean TPOT (Zeta)* **(j)** *P99 TPOT (Zeta)* **(k)** *P99.9 TPOT (Zeta)* **(l)** *Throughput (Zeta)*

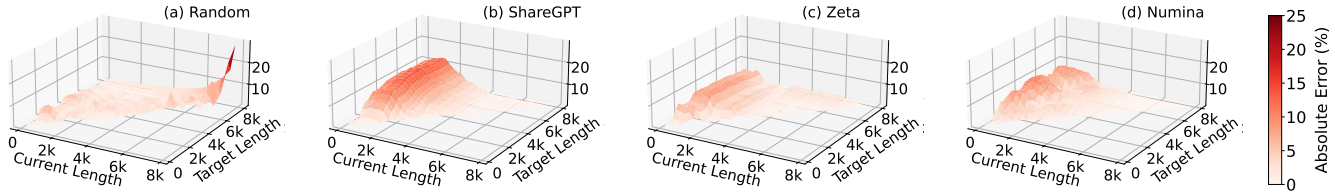
(m) *Mean TPOT (Numina)* **(n)** *P99 TPOT (Numina)* **(o)** *P99.9 TPOT (Numina)* **(p)** *Throughput (Numina)*

Figure 6. *End-to-end performance comparison among five methods across four datasets. TPOT is measured in milliseconds (ms), and throughput in tokens per second (tps).*

TPOT by 18.9%/31.5% on average against RR (with maxima of 29.2%/43.6%), by 16.2%/20.5% against ILS (max 29.3%/35.6%), and by 26.6%/38.0% against Migration, peaking at 48.2%/56.9%. The same pattern carries over to the heavy-tailed ShareGPT, the code-editing Zeta, and the reasoning-intensive Numina-Math traces: Sibyl's average P99.9 TPOT reductions over Migration reach 18.5%, 32.1%, and 30.6% respectively, with maxima ranging from 32.8% to 43.4%. Notably, Sibyl's lead over Migration is consistently larger than its lead over RR or ILS on every workload, indicating that reactive KV-cache transfer incurs substantial migration overhead and can even amplify tail latency under bursty contention, whereas proactive look-ahead avoids these costly corrections altogether.

Mean latency and throughput. The same direction holds for average latency and cluster capacity. Sibyl reduces Mean TPOT by 0.5% to 5.8% on average over RR and ILS, and by 1.0% to 9.0% over Migration, with the largest savings (9.0%/13.5% avg / max vs. Migration) appearing on Numina-Math, where chain-of-thought generation amplifies residual uncertainty and exposes the limits of reactive baselines. At the same time, throughput improves by up to 12.9% over RR, 10.4% over ILS, and 16.9% over Migration, showing that Sibyl suppresses tail latency without sacrificing cluster capacity.

How close is Sibyl to the upper bound? For this comparison, we define the reference baseline at each request rate as the mean of RR, ILS, and Migration on the metric of interest, and report the percentage improvement of Sibyl (or Oracle) over this baseline averaged and maximized across all tested rates. Under this aggregation, Oracle, which uses ground-truth decode lengths, improves Mean / P99 / P99.9 TPOT by 6.7%/22.3%/29.1% on average (with maxima of 11.8%/30.1%/37.4%), and throughput by 4.5% on average (7.4% at maximum). Sibyl, without any such oracle access, already attains 3.1%/15.6%/21.8% on the latency metrics and 3.3% on throughput. Translated into raw performance, Sibyl sits within 3.9% of Oracle on Mean TPOT, 8.9% on P99 TPOT, 10.7% on P99.9 TPOT, and only 1.1% on throughput, recovering the majority of Oracle's achievable improvement over the



(a) Random **(b)** ShareGPT **(c)** Zeta **(d)** Numina

Figure 7. *Calibration error of survival probability estimation.*

deployable baselines. This indicates that the survival-based estimator is sufficiently accurate for practical scheduling, and that the dominant source of remaining error is the fundamental variance of LLM generation lengths rather than the algorithm itself. We further evaluate Sibyl under different model and hardware configurations, with additional results provided in Appendix D.

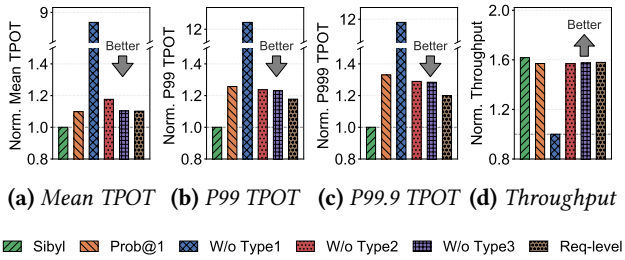
6.3 Estimation Error of Survival Probability

To validate the estimation error of Sibyl's profiling mechanism, we conduct the following analysis. We performe a static evaluation using the checkpoints trained on the Random, ShareGPT, Zeta and Numina datasets. Specifically, we queried the estimator on a dense grid of current lengths and target lengths across the entire trace, calculating the mean absolute error between the predicted survival probability $\mathcal{P}$ and ground-truth conditional probabilities derived from the cumulative distribution functions of the test datasets.

Figure 7 visualizes the error surface across the prediction space. Overall, Sibyl demonstrates robust calibration quality across diverse workload distributions. The Random workload shows negligible estimation error consistent with a uniform distribution, except for minor boundary effects at extreme context lengths. Even under the high output variance of ShareGPT, the estimator maintains an absolute error below 14%, with the vast majority of the configuration space exhibiting $< 5\%$ deviation. Similarly, for the coding-intensive Zeta dataset, the error remains under 9%. On the math-reasoning Numina dataset, whose CoT completions exhibit a heavy-tailed length distribution, the estimator achieves an absolute error below roughly 11%, with predictions becoming nearly exact once requests exceed ~2.5K context tokens.

6.4 Sensitivity Analysis

To validate Sibyl's design choices, we conduct a sensitivity analysis that selectively disables individual strategies or alters the load metric. Detailed methodology and variant definitions are provided in Appendix F. Figure 8 summarizes the results; for each metric, all values are normalized by the minimum value among all variants (lower is better for TPOT, higher is better for throughput).



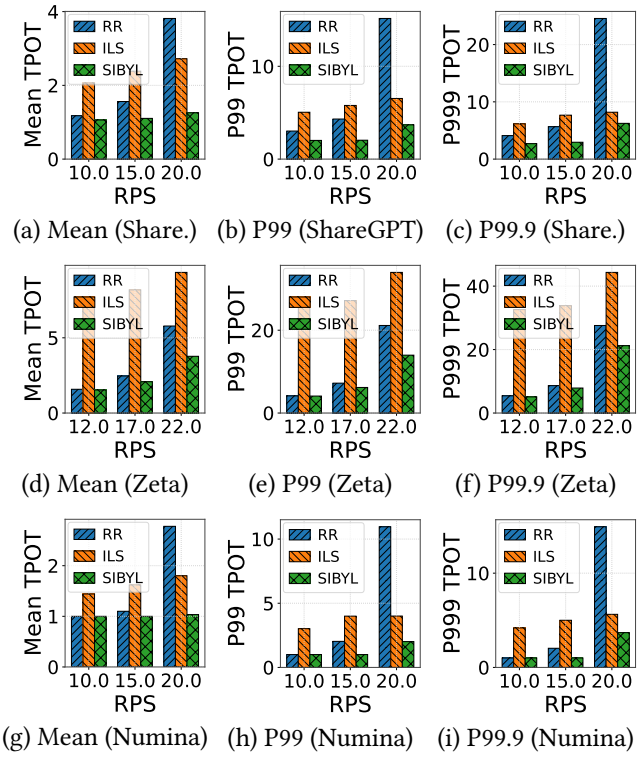
(a) *Mean TPOT* **(b)** *P99 TPOT* **(c)** *P99.9 TPOT* **(d)** *Throughput*

Figure 8. *Sensitivity analysis of Sibyl components on ShareGPT workloads.*

First, Prob@1 disables Sibyl's probabilistic survival weighting by fixing every request's survival probability to 1, i.e., optimistically assuming that all requests will keep running and thus keep contributing to decode load, rather than discounting them by their predicted survival probability. This inflates P99/P99.9 TPOT by 1.26×/1.33× and lowers throughput by 3%, confirming the importance of modeling survival uncertainty for tail latency.

Second, Sibyl estimates decode-phase load from three request categories: ongoing decode requests (Type 1), soon-to-start decode requests (Type 2), and post-handoff lookahead requests (Type 3). Removing Type 1 is the most damaging: P99 TPOT explodes to 12.3× (Mean 8.7×, P99.9 11.8×) and throughput collapses by 38%. Type 1 accounts for the active decoding tokens that constitute the dominant share of an instance's load; blind to it, the scheduler repeatedly routes new requests to already-saturated instances. Type 2 and Type 3 are likewise essential: dropping either still inflates P99.9 TPOT to 1.29× and 1.28× (with a ~3% throughput loss each), confirming that accurately accounting for soon-to-start and post-handoff requests is necessary to suppress worst-case interference. Together, the three categories form a complementary view of decode-phase load, and omitting any one of them measurably degrades tail latency.

Third, ReqLevel replaces our token-based load metric with simple request counts. This degrades P99/P99.9 TPOT



(a) Mean (Share.) (b) P99 (ShareGPT) (c) P99.9 (Share.)

(d) Mean (Zeta) (e) P99 (Zeta) (f) P99.9 (Zeta)

(g) Mean (Numina) (h) P99 (Numina) (i) P99.9 (Numina)

Figure 9. *Simulation on a cluster with 128 decode instances.*

by 1.18×/1.20× and reduces throughput by 2%, demonstrating that counting requests is too coarse to capture the heterogeneous token-level cost of LLM workloads.

6.5 Scalability Performance

To assess Sibyl's scalability in large-scale environments, we implemented a trace-driven discrete-event simulator modeling a cluster with 128 decode instances. The detailed methodology, along with extended evaluations on 256 and 512 instances, is provided in Appendix E.

Performance. Figure 9 shows the performance distribution. Sibyl consistently outperforms baseline strategies across all latency percentiles in large-scale settings. On the heavy-tailed ShareGPT workload, Sibyl improves Mean, P99, and P99.9 TPOT by 33.8%, 55.0%, and 56.4% compared to RR. The advantage over ILS is even more significant, yielding reductions of 52.9% in Mean, 60.7% in P99, and 56.5% in P99.9. The performance gap also shows on the Zeta dataset. Compared to RR, Sibyl reduces Mean, P99, and P99.9 TPOT by 18.1%, 21.9%, and 19.4%, respectively. Against ILS, Sibyl cuts Mean TPOT by 70.4% and suppresses tail latency by 78.0% (P99) and 77.7% (P99.9). The reasoning-driven Numina workload, whose long and highly variable output lengths inflate decode-side contention, exhibits the same trend: Sibyl lowers Mean, P99, and P99.9 TPOT by 19.7%, 42.5%, and 47.7% over RR, and by 37.8%, 71.1%, and 70.1% over ILS, with the tail-latency advantage widening as the output-length variance grows. These results illustrate the scalability bottlenecks of baselines. RR's blind scheduling and ILS's "invisible load" issue lead to significant performance degradation at scale. In contrast, Sibyl's look-ahead scheduling effectively mitigates these inefficiencies.

Dataset	RR	ILS	Sibyl
ShareGPT	53.7%	43.2%	**89.8%** (↑1.67×)
Zeta	36.8%	14.5%	**47.5%** (↑1.29×)
Numina	67.1%	67.4%	**99.9%** (↑1.48×)

Table 4. *Optimal assignment ratio comparison.*

Optimal Assignment Ratio. We further analyze the placement accuracy, defined as the probability that the pre-assigned instance remains the least-loaded node at the actual handoff moment. Table 4 shows that ILS can underperform RR under long-context or highly dynamic workloads (*e.g.,* 14.5% vs. 36.8% on Zeta), indicating that stale load snapshots may cause concurrent requests to collide on the same seemingly idle instance. In contrast, Sibyl consistently improves assignment precision across all datasets, achieving 89.8% on ShareGPT, 47.5% on Zeta, and 99.9% on Numina. Compared with the stronger baseline in each dataset, Sibyl improves the optimal assignment ratio by up to 1.67×, and more than doubles ILS's precision on Zeta. The relatively lower accuracy on Zeta is due to its longer code inputs, which extend the prefill phase, widen the prediction window, and increase prediction uncertainty. This improved assignment precision directly explains the latency reductions observed in Figure 9.

6.6 Overhead Analysis

We analyze the runtime overhead of Sibyl along two dimensions: the scheduling decision latency and the computational resource consumption incurred on the proxy node.

Decision Latency. Table 5 reports the average time to dispatch a single request. Whereas lightweight policies such as RR and ILS make decisions almost instantaneously (< 1 ms), Sibyl incurs an average latency of 5.07 ms, the bulk of which stems from the prefill-duration prediction model used to estimate the handoff moment. Nevertheless, since end-to-end inference latencies for LLM workloads typically span seconds to minutes, this millisecond-level overhead is orders of magnitude smaller and thus negligible to the end-user experience.

Computational Cost. As the scheduling logic for all methods runs on the CPU-based proxy, we track CPU utilization to assess efficiency. As shown in Figure 10, Sibyl yields a utilization profile closely matching the baselines, staying within a low range (2.5%–4.5%) with no noticeable spikes. This confirms that the probabilistic computation and state tracking in Sibyl are lightweight and impose no computational bottleneck on the serving system.

Method	Latency (ms)
RR	<1
ILS	<1
SIBYL	5.07

Table 5. *Comparison of scheduling decision latency.*

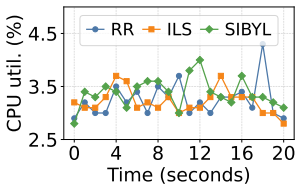


Figure 10. *CPU utilization of the scheduler proxy over time.*

7 Discussion

Learning-based Length Prediction. Recent works explore learning-based approaches that predict generation length to assist request placement [11, 17, 19, 53]. While effective in controlled settings, these designs introduce additional prediction overhead on the scheduling path and require invoking a model for every request. Moreover, learned length prediction faces a robustness–overhead trade-off: higher accuracy typically costs more, while lightweight predictors are less reliable across workloads.

Robustness to Prediction Errors. SIBYL relies on lightweight profiling primitives, including coarse-grained decoding rates (v_k) and survival probabilities ($\mathcal{P}_k$). Importantly, these estimates are not required to be exact: the scheduler only depends on their relative ordering across instances, rather than precise numerical values. As a result, SIBYL can tolerate moderate estimation inaccuracies, as long as the projected load preserves the relative pressure across candidates. This design avoids the need for tightly calibrated predictors on the critical scheduling path.

Limitations and Future Work. (1) The effectiveness of SIBYL is workload-dependent. When decoding is short, when prefill dominates latency, or when decode resources are lightly loaded, the load discrepancy between scheduling and execution is limited, leaving less headroom for projection-based early assignment [9]. (2) SIBYL adopts a per-request greedy strategy to enable fast and scalable scheduling decisions. While global optimization across concurrent arrivals may achieve better assignments under bursty workloads, it would incur higher coordination and computation overhead. (3) SIBYL focuses on mitigating load imbalance at the instance level. It does not explicitly address intra-instance imbalance, such as GPU-level skew introduced by expert parallelism [5, 8, 52, 63]. Incorporating finer-grained, device-level signals into load projection is a promising direction for future work.

8 Related Work

Disaggregated LLM serving. Motivated by the contrasting resource profiles of prefill and decode, disaggregated serving places the two phases on separate instance pools to remove interference and scale each phase independently [7, 18, 31, 35, 36, 46, 65]. Subsequent systems disaggregate additional stages [25, 44], relocate the boundary to trade TTFT for TPOT [14, 27], pool stranded memory [15], make instances stateless to flip roles under bursty traffic [38, 56], separate prefill and decode *within* a single GPU [42], or unify aggregated and disaggregated execution [58]. Across this spectrum, the architecture defines *where* the phases execute, yet shares a common premise: a request is bound to its decode instance at prefill start so that KV-cache transfer overlaps with computation [35, 36]. SIBYL is orthogonal: it leaves the architecture untouched and supplies the missing primitive of *which* decode instance to bind to, dropping in as a black-box proxy for any such system.

Predictive and length-aware scheduling. Since output length is unknown a priori, engines often fall back on first-come-first-serve and suffer head-of-line blocking. A growing line predicts request lengths and orders execution accordingly: S^3 [19] packs batches from predicted lengths, learning-to-rank approximates shortest-job-first from predicted output ranks [11], and embedding-based [40], iterative [3], and semi-clairvoyant [22] predictors drive shortest-remaining-time scheduling within a single engine. Yet exact per-request prediction is infeasible [11] and brittle under the heavy-tailed outputs of reasoning workloads (§2.2), and these methods target intra-instance *ordering* rather than placement. SIBYL instead decides cross-instance *placement* without point estimates: it models active-task *residual lifetimes* via survival analysis and incorporates shadow load from in-flight prefills, yielding a probabilistic occupancy projection.

Capacity provisioning and autoscaling. A complementary direction adapts the *amount* of provisioned capacity to demand: autoscaling that cuts scaling latency via live layer-granular scaling over the compute network [61] or low-cold-start serverless inference [10], energy co-optimization [45], coordination of heterogeneous disaggregated pools [23], and cross-model GPU pooling [57]. Closest to us, Aladdin [30] jointly scales resources and places requests by bin-packing on predicted lengths. These mechanisms reconfigure the cluster over coarse timescales; SIBYL is orthogonal and operates at per-request granularity on a *fixed* pool, complementing such elasticity after capacity provisioning.

9 Conclusion

We identify a fundamental *load discrepancy* in disaggregated LLM serving, where scheduling decisions made at prefill start often become stale by the time decoding begins. To address this issue, we present SIBYL, a speculative scheduler that projects per-instance load at the handoff moment using lightweight survival-based primitives. This projection guides load-aware early assignment and helps align early assignment with actual decode execution. Experiments on production traces and large-scale simulations show that SIBYL reduces tail latency by up to 56.9% compared to state-of-the-art schedulers.

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APPENDIX

A Why Tail Latency Matters?

In production LLM serving, relying solely on central tendency metrics (such as Mean or P50) often creates a false sense of security regarding system performance. Figure 1 illustrates a typical latency distribution observed under standard scheduling policies. While the median (P50) TPOT comfortably satisfies the stringent SLO of 100ms (indicated by the red dashed line), the distribution exhibits a severe long-tail phenomenon. As shown, the tail latencies degrade exponentially: the P99 latency exceeds 1500ms, and the P99.9 latency spikes to over 2500ms, more than **25× higher** than the SLO threshold. For interactive applications like chatbots, these tail latencies manifest as noticeable "stuttering" during text generation, severely disrupting the user experience. Consequently, optimizing for tail latency (P99/P99.9) rather than P50 is critical for ensuring consistent quality of service, motivating the design of Sibyl to specifically mitigate these worst-case scenarios.

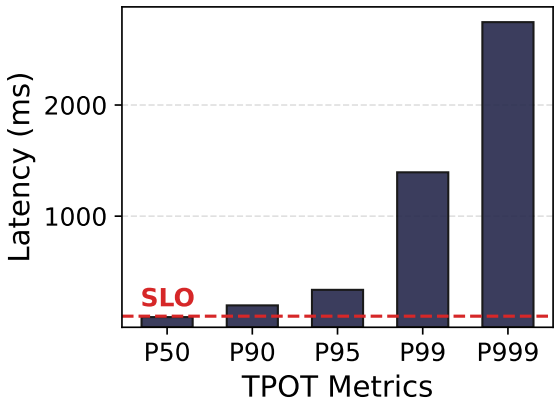


Figure 1. The long-tail latency problem. *Although the median (P50) TPOT meets the 100ms SLO (red dashed line), severe congestion causes P99 and P99.9 latencies to violate the constraint by orders of magnitude, highlighting the inadequacy of P50 as a sole metric.*

B Temporal Consistency

Our formulation is designed to be *boundary-consistent* across state transitions to ensure scheduling stability. First, at the *Active-Shadow boundary* (Type 1 ↔ Type 2), a task transitioning from queue to execution shifts from using the system-average speed ($\bar{v}_{sys}$) to its specific rate (v_k). Mathematically, since v_k is initially unknown, $\bar{v}_{sys}$ serves as its unbiased estimator at the boundary. While the realization of v_k during execution may introduce a value update, the *expected load* remains statistically consistent at the instant of transition.

Second, at the *Shadow-Decay boundary* (Type 2 ↔ Type 3), the model naturally converges: as the projected start time aligns with the handoff ($\tau_k \to \tau$), the generation component vanishes, and both Type 2 and Type 3 models resolve identically to the intrinsic input cost ($\lim_{\tau_k \to \tau} \mathcal{L} = l_k^{in}$). This design eliminates singularity points in the objective function, allowing the scheduler to smoothly track tasks as they drift through different lifecycle stages.

Table 1. *Generation hyperparameters used to enable reasoning mode.*

Parameter	Value
Temperature	0.6
Top-P	0.95
Top-K	20
Min-P	0
Greedy Decoding	Disabled

C Detailed Experimental Setup

C.1 Model Configurations

To enable the reasoning capabilities of Qwen3-8B and avoid deterministic degeneration, we disable greedy decoding and employ stochastic sampling. We adopt the official hyperparameters recommended by Qwen Team [37], as detailed in Table 1.

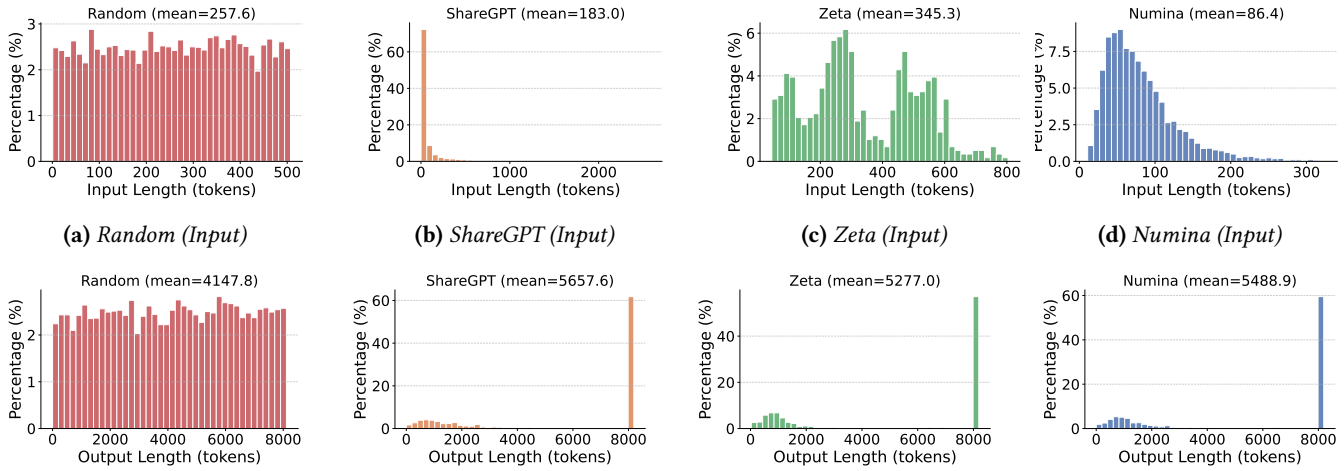
C.2 Dataset Details

We evaluate our system using four datasets that cover synthetic micro-benchmarks, real-world conversations, code generation tasks, and mathematical reasoning. The input and output length distributions are visualized in Figure 2.

Random. This synthetic dataset is designed to micro-benchmark system performance under reasoning-intensive scenarios (*e.g.,* Chain-of-Thought), where short questions trigger long responses. We uniformly sample input prompt lengths from $[1, 512]$ and output lengths from $[1, 8192]$. This configuration stresses the decode phase and allows us to test the scheduler's behavior under uniform workload variance.

ShareGPT. Representing real-world human-chatbot interactions, this dataset features relatively short prompts (mean length 183) but exhibits high variance in response lengths. Under our reasoning-enabled configuration, the output distribution becomes extremely heavy-tailed; as shown in Figure 2(e), about 62% of requests saturate the maximum token limit (8192), creating sustained pressure on decode instances.

Zeta. Sourced from predictive code editing tasks, this dataset introduces longer context prefixes compared to S hareGPT, with input lengths extending up to 1.7k tokens (mean 345). Similar to ShareGPT, the output distribution is dominated by long-running tasks that frequently hit the generation cap (57% reach 8192 tokens), effectively testing the system's ability to handle heavier prefill loads alongside concurrent long-context decoding.



(a) *Random (Input)* **(b)** *ShareGPT (Input)* **(c)** *Zeta (Input)* **(d)** *Numina (Input)*

(e) *Random (Output)* **(f)** *ShareGPT (Output)* **(g)** *Zeta (Output)* **(h)** *Numina (Output)*

Figure 2. Token length distributions across datasets. *The top row shows the input prompt length distributions, while the bottom row shows the output generation length distributions for Random, ShareGPT, Zeta, and Numina.*

Numina. Drawn from competition-level mathematical problems, this dataset pairs concise problem statements (mean input length 86) with lengthy step-by-step reasoning traces. The output distribution is the most decode-heavy among all datasets, with an average response length of roughly 5.5k tokens and nearly 60% of requests exhausting the 8192-token budget, making it a stringent test for sustained decode throughput under minimal prefill cost.

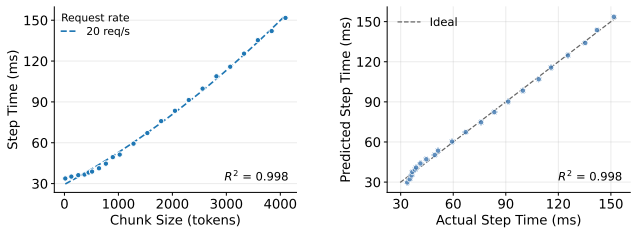
```
# Execution Command for Prefill Instance
vllm serve <MODEL_PATH> \
    --host <HOST_IP> \
    --port <HTTP_PORT> \
    --served-model-name qwen3-8b \
    --tensor-parallel-size 1 \
    --seed 1024 \
    --dtype float16 \
    --max-model-len 32768 \
    --max-num-batched-tokens 32768 \
    --max-num-seqs 1 \
    --trust-remote-code \
    --reasoning-parser qwen3 \
    --gpu-memory-utilization 0.9 \
    --compilation-config '{"cudagraph_mode": "PIECEWISE"}' \
    --kv-transfer-config '{
        "kv_connector": "P2pNcclConnector",
        "kv_role": "kv_producer",
        "kv_buffer_size": "1e1",
        "kv_port": "<KV_PORT>",
        "kv_connector_extra_config": {
            "proxy_ip": "<PROXY_IP>",
            "proxy_port": "<PROXY_PORT>",
            "http_port": "<HTTP_PORT>"
        }
    }'
```

```
# Execution Command for Decode Instance
vllm serve <MODEL_PATH> \
    --host <HOST_IP> \
    --port <HTTP_PORT> \
    --served-model-name qwen3-8b \
    --tensor-parallel-size 1 \
    --seed 1024 \
    --dtype float16 \
    --max-model-len 32768 \
    --max-num-batched-tokens 32768 \
    --max-num-seqs 256 \
    --trust-remote-code \
    --reasoning-parser qwen3 \
    --gpu-memory-utilization 0.8 \
    --kv-transfer-config '{
        "kv_connector": "P2pNcclConnector",
        "kv_role": "kv_consumer",
        "kv_buffer_size": "8e9",
        "kv_port": "<KV_PORT>",
        "kv_connector_extra_config": {
            "proxy_ip": "<PROXY_IP>",
            "proxy_port": "<PROXY_PORT>",
            "http_port": "<HTTP_PORT>"
        }
    }'
```

```
# Benchmarking Command
vllm bench serve \
    --backend vllm \
    --model qwen3-8b \
    --tokenizer <MODEL_PATH> \
    --dataset-name "hf" \
    --dataset-path <DATASET_PATH> \
    --hf-name <DATASET_NAME> \
    --host <PROXY_IP> \
    --port <PROXY_PORT> \
    --burstiness <BURSTINESS> \
    --percentile-metrics "ttft,tpot,itl,e2el" \
    --metric-percentiles "50,90,95,99,99.9" \
    --seed $(date +%s) \
    --trust-remote-code \
    --request-rate <RPS> \
    --num-prompts <NUM_PROMPTS>
```

C.3 vLLM Runtime Configuration

We deploy the disaggregated inference system using vLLM. Specifically, we configure the system with the `P2pNcclConnector`, which natively supports the overlapping of prefill computation with KV cache transmission. The following listings provide the representative command-line templates used in our experiments. Environment-specific variables (*e.g.*,



(a) *Step time vs. chunk size* **(b)** *Actual vs. predicted*

Figure 3. Offline fitting of the prefill per-step latency. *(a) Per-step latency scales smoothly with chunk size and is fit by a low-degree polynomial ($R^2 = 0.998$). (b) Predicted step time closely matches the measured values.*

IPs, ports) and dynamic workload parameters have been replaced with placeholders for generality.

C.4 Prefill Step-Time Fitting

As described in §5, Sibyl estimates a request's prefill completion time from its number of chunked-prefill steps and a per-step latency model profiled offline on the target hardware. Figure 3(a) shows that, with `batch_size=1` and chunked prefill, the per-step latency grows smoothly with the chunk size and is well captured by a low-degree polynomial ($R^2 = 0.998$). Figure 3(b) confirms that the fitted model closely tracks the measured step time across the operating range. This lightweight model lets the proxy translate a prompt length into an accurate handoff-time estimate at negligible cost.

D Additional Experiment Results

The primary evaluation in the main text utilizes NVIDIA H800 GPUs. To validate the broad applicability and robustness of Sibyl across diverse hardware architectures, this section supplements those results with experiments on two additional platforms (Table 2): the NVIDIA H20, a decode-optimized accelerator with massive HBM capacity (141 GB) and high bandwidth (4.0 TB/s) but relatively lower compute power (148 TFLOPS, BF16), making it exceptionally suitable for the memory-bound decode phase of LLM inference; and the NVIDIA A100, a flagship predecessor with significantly higher compute capability (312 TFLOPS, BF16) but more constrained memory resources (40 GB HBM, 1.89 TB/s).

Table 2. *Hardware specification comparison among NVIDIA H20, A100, and H800 GPUs.*

Specification	NVIDIA H20	NVIDIA A100	NVIDIA H800
FP16/BF16 Compute	148 TFLOPS	312 TFLOPS	1,979 TFLOPS
HBM Capacity	141 GB	40 GB	80 GB
HBM Bandwidth	4.0 TB/s	1.89 TB/s	3.35 TB/s

D.1 Experimental Setup

NVIDIA H20. We deploy Sibyl on a server node with eight NVIDIA H20 GPUs and evaluate Qwen3-32B [59], which fits within the H20's 141 GB HBM without tensor parallelism [43]. To accommodate the varying prefill–decode ratios of different workloads, we adopt a 2P4D topology (2 prefill and 4 decode instances) for the Random dataset and a 4P4D topology for the ShareGPT and Zeta datasets, with each GPU hosting a single model replica.

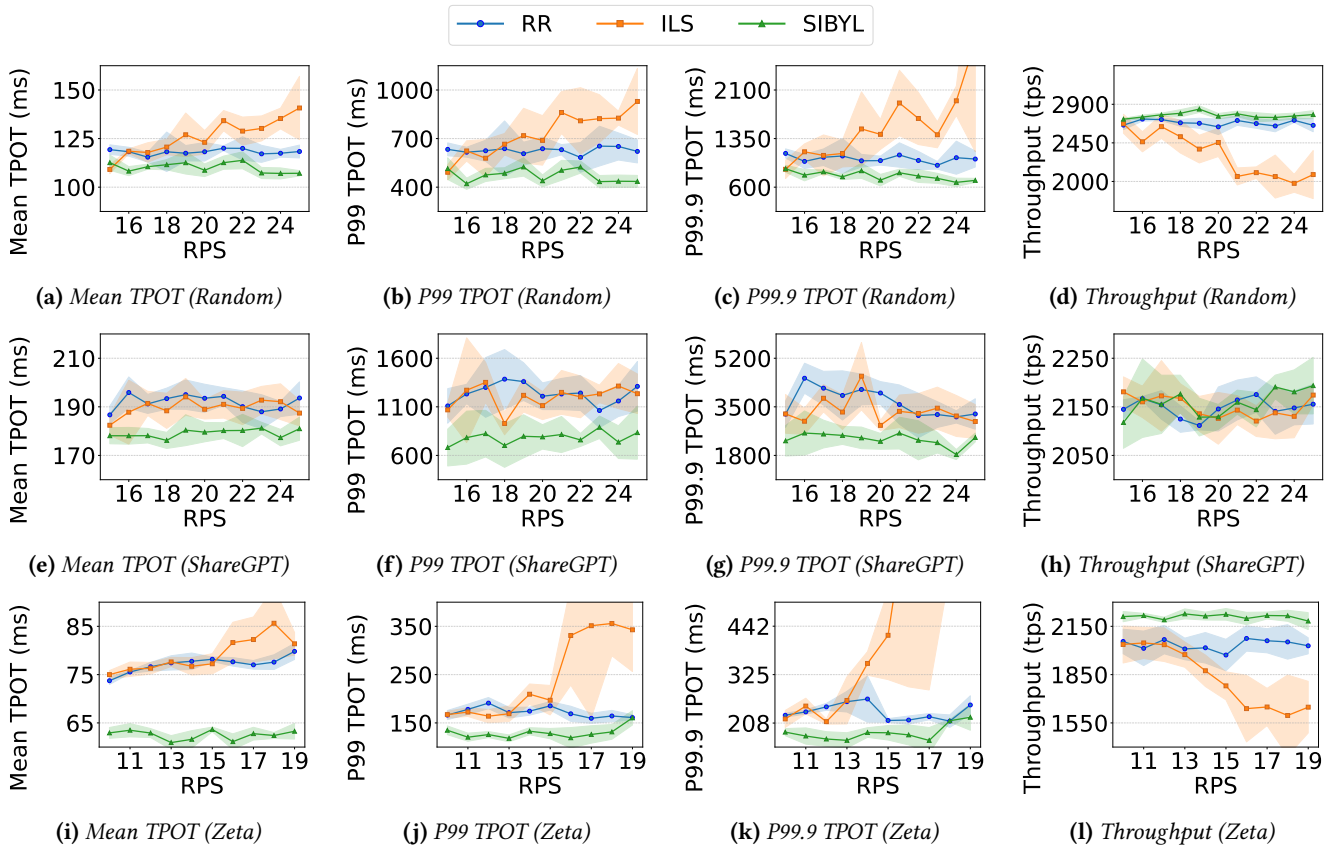
NVIDIA A100. We further conduct experiments on a distributed cluster of two physical servers equipped with NVIDIA A100 GPUs, deployed in a 2P4D topology where the prefill workload runs on a 2×A100 server and the decode workload on a separate 4×A100 server, with each GPU hosting a single model replica without tensor parallelism. Here we evaluate two widely adopted open-weights models, Qwen2.5-7B-Instruct [50] and Llama3-8B [2], on a synthetic random dataset whose input prompt lengths are uniformly distributed in $[1, 1024]$ and output generation lengths in $[1, 4096]$.

Across both platforms, we compare Sibyl against the RR and ILS baselines, mirroring the deployable-scheduler comparison of the main text.

D.2 Overall Performance

D.2.1 NVIDIA H20. Figure 4 summarizes the end-to-end performance of Sibyl, ILS, and RR with Qwen3-32B across three representative workloads on the H20 platform. First, on the bursty Random workload, Sibyl increases output-token throughput by 1.21× (vs. ILS) and 1.03× (vs. RR), while reducing Mean TPOT by 12.0% (vs. ILS) and 6.7% (vs. RR). Notably, it effectively suppresses tail latency, lowering P99 and P99.9 TPOT by 32.7% and 43.4% (vs. ILS), as well as by 24.5% and 25.2% (vs. RR). Second, on the real-world ShareGPT dataset, while throughput remains comparable across methods due to system saturation (∼1.0×), Sibyl stabilizes generation performance by mitigating hotspots: it reduces Mean TPOT by ∼6% and cuts tail latency drastically, lowering P99 TPOT by 34.3% (vs. ILS) and 36.3% (vs. RR), and P99.9 TPOT by 28.6% (vs. ILS) and 34.8% (vs. RR). Finally, on the Zeta dataset, Sibyl achieves its most pronounced gains, boosting throughput by 1.22× (vs. ILS) and 1.09× (vs. RR) and reducing Mean TPOT by 20.7% (vs. ILS) and 18.9% (vs. RR). More importantly, Sibyl achieves strict SLO compliance with massive reductions in tail latency, where P99 and P99.9 TPOT drop by 41.9% and 50.6% (vs. ILS), as well as 24.2% and 20.8% (vs. RR). These results collectively show that Sibyl's predictive scheduling not only maximizes cluster capacity in bursty scenarios but also ensures prompt responsiveness in production-grade workloads.

D.2.2 NVIDIA A100. Figure 5 illustrates the end-to-end performance comparison on the random dataset. We observe

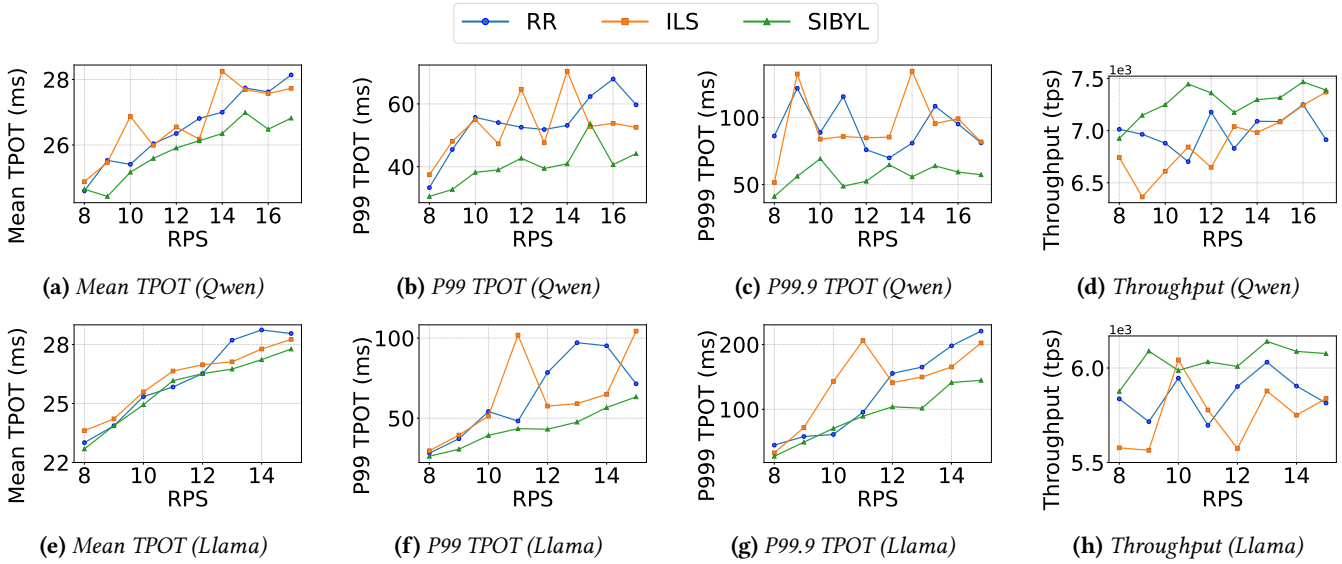


(a) *Mean TPOT (Random)* **(b)** *P99 TPOT (Random)* **(c)** *P99.9 TPOT (Random)* **(d)** *Throughput (Random)*

(e) *Mean TPOT (ShareGPT)* **(f)** *P99 TPOT (ShareGPT)* **(g)** *P99.9 TPOT (ShareGPT)* **(h)** *Throughput (ShareGPT)*

(i) *Mean TPOT (Zeta)* **(j)** *P99 TPOT (Zeta)* **(k)** *P99.9 TPOT (Zeta)* **(l)** *Throughput (Zeta)*

Figure 4. *End-to-end performance comparison of* Sibyl*, ILS, and RR with Qwen3-32B [59] on NVIDIA H20 across the Random, ShareGPT, and Zeta workloads.*



(a) *Mean TPOT (Qwen)* **(b)** *P99 TPOT (Qwen)* **(c)** *P99.9 TPOT (Qwen)* **(d)** *Throughput (Qwen)*

(e) *Mean TPOT (Llama)* **(f)** *P99 TPOT (Llama)* **(g)** *P99.9 TPOT (Llama)* **(h)** *Throughput (Llama)*

Figure 5. *End-to-end performance comparison of Qwen2.5-7B-Instruct [50] and Llama3-8B [2] on random dataset.*

that Sibyl consistently outperforms both RR and ILS baselines across different models and varying request rates (RPS), particularly in suppressing tail latency and improving system throughput.

Qwen2.5-7B Results. As shown in the top row of Figure 5, SIBYL demonstrates robust stability under high contention. Compared to ILS, SIBYL reduces the average P99 and P99.9 TPOT by **23.0%** and **36.2%**, respectively, with peak reductions reaching up to **58.6%** in the most congested scenarios. Similarly, against RR, SIBYL achieves average reductions of **24.2%** (P99) and **36.4%** (P99.9). In terms of throughput, SIBYL improves the average throughput by **5.7%** over ILS and **4.2%** over RR, effectively handling higher loads without violating latency constraints.

Llama3-8B Results. The bottom row of Figure 5 confirms the generalizability of our approach. SIBYL achieves significant tail latency improvements, reducing the average P99 TPOT by **26.4%** vs. ILS and **26.2%** vs. RR. For the strict P99.9 metric, SIBYL lowers the latency by an average of **32.0%** compared to ILS. Throughput gains remain consistent, with SIBYL outperforming ILS by an average of **5.1%** and RR by **3.1%**.

While the improvement in Mean TPOT is modest (approximately 2–3% across workloads), the substantial reduction in tail latency (P99/P99.9) indicates that SIBYL effectively mitigates the severe congestion instances caused by suboptimal scheduling. Crucially, this suppression of tail latency directly translates to significantly fewer SLO violations, thereby guaranteeing a higher quality of service.

E Simulation Methodology

To evaluate SIBYL at scales exceeding typical physical testbeds (e.g., 64–256 decode instances) and to conduct extensive stress testing under varied arrival rates, we implemented a high-fidelity, trace-driven discrete-event simulator. This section details the simulation environment, the hardware modeling assumptions, and the implementation logic corresponding to Algorithms 2 and 3.

E.1 Hardware Modeling

Unlike basic simulators that assume constant processing speeds, our model explicitly accounts for memory bandwidth contention. Since LLM decoding is bandwidth-bound, we model the GPU as a shared resource where the total throughput is divided among active requests. Consequently, as the number of concurrent tasks increases, the execution speed of each individual request decreases, accurately reflecting the performance slowdown in real hardware.

Dynamic Throughput. The total system throughput (TPS_{total}) of a decode instance is not static; it exhibits nonlinear variations governed by the request concurrency (N) and the total token volume (T). Instead of relying on simple univariate polynomial fitting, we accurately model these hardware dynamics using a *Random Forest Regressor* trained on profiling data from the NVIDIA H20 GPU (visualized in Figure 6). The model predicts throughput as the ensemble

Algorithm 2: Trace-Driven Discrete-Event Simulation Framework

Input : Request Trace $\mathcal{T}$, Cluster Config C, Scheduler $\mathcal{S}$
Output : Latency Metrics, Throughput

Initialize N Decode instances, global `RequestTable` (for SIBYL), and set $t_{now} \leftarrow 0$;
for *each request* r *in* $\mathcal{T}$ *sorted by arrival time* **do**
  $t_{now} \leftarrow r.arrival\_time$;
  for *each instance* $j \in \{1 \dots N\}$ **do**
    AdvanceInstance(j, t_{now}) ; // See Algorithm 3
    Update instance speed v_j based on dynamic load;
  if $\mathcal{S}$ *is SIBYL* **then**
    Sync `RequestTable` with physical state;
    $\delta_{pre} \leftarrow$ EstimatePrefill$(r.input_len)$;
    $\tau \leftarrow t_{now} + \delta_{pre}$;
    $j^* \leftarrow \arg\min_j$ ProjectedLoad(j, τ);
  else if $\mathcal{S}$ *is ILS* **then**
    $j^* \leftarrow \arg\min_j$ CurrentLoad(j, t_{now});
  else
    $j^* \leftarrow (j_{last} + 1) \pmod N$;
  $t_{start\_decode} \leftarrow t_{now} + \delta_{pre}$;
  Push event `START_DECODE(r)` to instance j^* queue at t_{start_decode};
Flush all instances until completion;

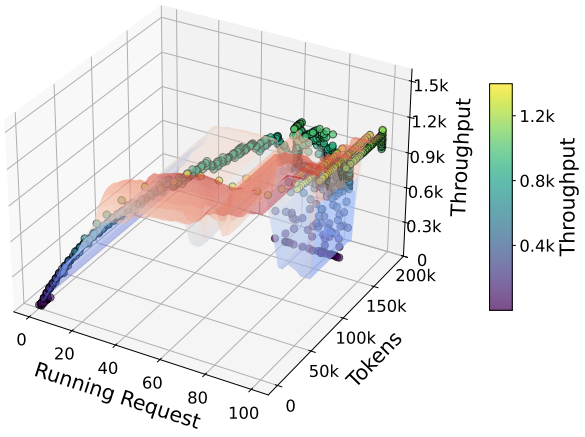


Figure 6. *TPS profiling on NVIDIA H20 GPU.*

average of 100 decision trees:

$$TPS_{total}(N, T) = \frac{1}{100} \sum_{i=1}^{100} \text{Tree}_i(N, T) \tag{18}$$

Evaluation on a hold-out test set demonstrates high fidelity with an R^2 of 0.9024. Feature importance analysis reveals that

while concurrency is the dominant factor (85.51%), the total token load exerts a significant influence (14.49%), validating the necessity of a bi-variate model to capture memory bandwidth saturation effects. In our simulation (Algorithm 3, Line 4), we query this inference model to dynamically recalibrate instance speed based on the real-time state (N, T).

Bandwidth Contention. When multiple requests decode concurrently, they share the GPU's memory bandwidth. The simulator models this by splitting the total throughput equally among all active tasks (Algorithm 3, Line 5). Consequently, the effective speed v_{speed} of a single request decreases as the batch size increases, accurately simulating the slowdown effect caused by multiple parallel requests.

E.2 Simulation Logic and Workflow

The simulation proceeds as a strict discrete-event system driven by the request arrival trace. The interaction between the high-level scheduling logic and the low-level physics engine is detailed below:

1. Global Event Loop (Algorithm 2). The framework processes requests sequentially based on their arrival times. Upon each arrival at time t_{now}, the system first ensures state synchronization (Lines 6-9) by invoking the AdvanceInstance subroutine for every instance; this advances the physical simulation to the current timestamp, processing any interim task completions or new decoding starts. With the system state synchronized, the scheduler then executes the decision-making logic (Lines 11-17) to select a target instance j^*. For Sibyl, this involves syncing the shadow `RequestTable` and projecting loads at the future handoff time τ. Crucially, to explicitly model the temporal gap (Lines 19-20), the request is not executed immediately; instead, a `START_DECODE` event is registered in the target instance's queue, scheduled to trigger strictly after the prefill duration at $t_{now} + \delta_{pre}$.

2. Instance State Evolution (Algorithm 3). This subroutine acts as the physics engine. It advances the state of a specific instance continuously. Instead of fixed time steps, it calculates Δt_{step} (Line 9) as the minimum time until the next discrete event: either a running task finishes its generation (Δt_{finish}) or a pending prefill task enters the decode phase (Δt_{start}). This event-driven progression ensures that resource contention is recalculated exactly when the system load changes, guaranteeing high temporal precision.

Algorithm 3: Instance State Evolution (Processor Sharing Model)

1 **Function** *AdvanceInstance*$(instance, t_{target})$
  // Simulates continuous execution up to t_{target}
2 **while** $instance.time < t_{target}$ **do**
3 $Q_{run} \leftarrow instance.decoding\_tasks$;
4 $L_{tokens} \leftarrow \sum_{k \in Q_{run}} (len_{prompt,k} + len_{generated,k})$;
5 $TPS_{total} \leftarrow$ GetFittedThroughput(Q_{run}, L_{tokens}) ; // Based on profiling curve (Figure 6)
6 $v_{speed} \leftarrow TPS_{total}/|Q_{run}|$; // Processor sharing: split equally
  // Determine time to next state change
7 $\Delta t_{finish} \leftarrow \min_{k \in Q_{run}} (remaining\_tokens_k / v_{speed})$;
8 $\Delta t_{start} \leftarrow instance.next\_start\_event - instance.time$;
9 $\Delta t_{step} \leftarrow \min(\Delta t_{finish}, \Delta t_{start}, t_{target} - instance.time)$;
  // Advance physical state
10 **for** *each task* k *in* Q_{run} **do**
11 $remaining\_tokens_k \leftarrow remaining\_tokens_k - (\Delta t_{step} \times v_{speed})$;
12 $instance.time \leftarrow instance.time + \Delta t_{step}$;
  // Handle discrete events
13 **if** $\Delta t_{step} == \Delta t_{finish}$ **then**
14 Identify task k^* with $remaining \leq 0$;
15 Remove k^* from Q_{run}, record t_{end} and TPOT;
16 **else if** $\Delta t_{step} == \Delta t_{start}$ **then**
17 Pop task r_{new} from event queue;
18 Add r_{new} to Q_{run} with initial remaining load;

F Definitions of Sensitivity Variants

In this section, we provide precise definitions for the variants evaluated in the sensitivity analysis (Section 6.4). To isolate the contribution of each modeling component in Sibyl, we modify the objective function or the load estimation logic as follows:

Prob@1 (No Probabilistic Weighting): This variant disables the survival analysis module. Instead of computing the conditional survival probability based on historical distributions, we assume a deterministic worst-case scenario where every active task persists until the completion of the incoming request's prefill phase. Mathematically, we set the probability term $\mathcal{P} = 1$ in Eq. (15) and Eq. (16). This baseline evaluates the necessity of risk-adjusted load estimation versus conservative over-provisioning.

w/o Type 1 (No Active Task Projection): We exclude the projected load from currently running active tasks by setting $\mathcal{L}_{\mathrm{I}}(k, \tau) = 0$. This variant forces the scheduler to ignore the residual workload on decode instances, relying solely on the

estimated pressure from pending (shadow) tasks. This tests whether knowing the state of currently executing requests is critical for decision-making.

w/o Type 2 (No Pre-Handoff Shadow): We ignore the "invisible" contention from requests that are currently in the prefill phase but are projected to start decoding *before* the target request arrives ($\tau_k \leq \tau$). By setting $\mathcal{L}_{\text{II}}(k, \tau) = 0$, the scheduler fails to account for the queue that will form during the prefill interval. This baseline highlights the impact of the temporal gap described in §2.2.

w/o Type 3 (No Post-Handoff Shadow): We exclude the interference decay model for tasks projected to start *after* the handoff moment ($\tau_k > \tau$). By setting $\mathcal{L}_{\text{III}}(k, \tau) = 0$, the scheduler optimizes only for the expected load at moment τ, ignoring imminent future arrivals that might cause congestion shortly after.

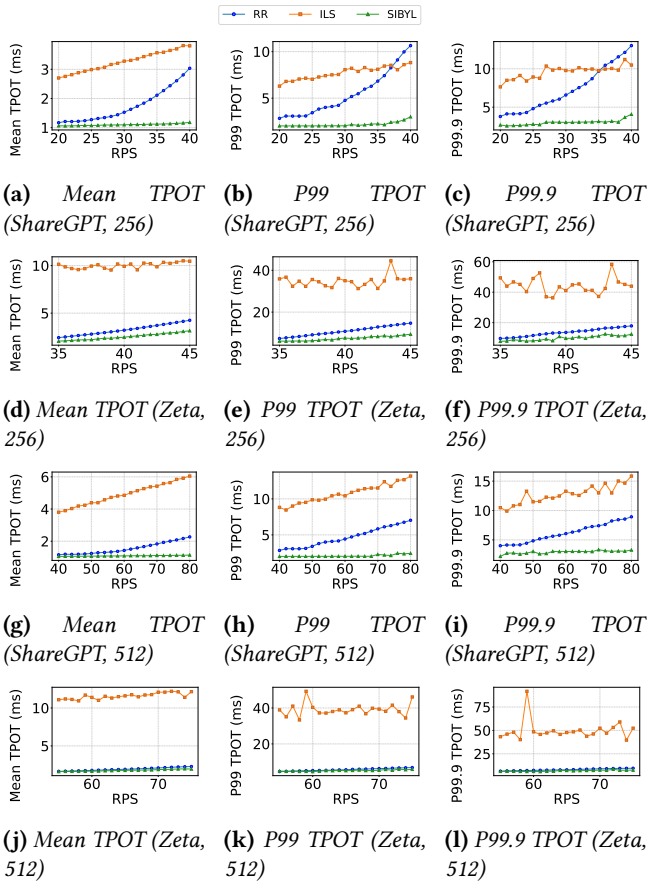
Req Level (Request-Level Balancing): This variant abandons the token-based load metric entirely. Instead of modeling physical memory pressure (token occupancy) and duration, it treats all requests as identical units. The scheduler assigns the new request to the instance with the fewest total tasks, regardless of their generation lengths or memory footprint. Notably, unlike the ILS baseline, this variant explicitly accounts for the count of pending requests currently in the prefill phase. This serves as a baseline to validate the necessity of fine-grained, token-aware modeling in heterogeneous LLM workloads.

G Extended Simulation Results

To verify the scalability of Sibyl in massive large-scale environments, we extend our simulation to clusters with 256 and 512 decode instances. Figure 7 presents the latency characteristics under sweeping arrival rates.

Latency Stability at Scale. Sibyl demonstrates robust scalability, maintaining a flat latency profile even as the system approaches saturation. In contrast, the benchmark methods suffer from varying degrees of performance degradation. On the heavy-tailed ShareGPT workload, RR blindly routes long requests to busy instances, causing unbalanced load. Sibyl effectively suppresses this tail latency, outperforming RR by 54.33% (256 instances) and 51.66% (512 instances) in P99 TPOT, while surpassing ILS by 71.73% and 80.21% respectively. On the Zeta dataset, the gap widens further as ILS struggles with long prefill durations that create large temporal gaps. This "invisible load" issue renders ILS's instantaneous view obsolete, causing severe congestion. Consequently, Sibyl achieves drastic reductions in P99 TPOT against ILS: 78.65% on 256 instances and 86.66% on 512 instances, confirming that predictive lookahead is critical for scaling coding workloads.

Mechanistic Explanation via Assignment Accuracy. Table 3 elucidates the root cause of these performance gaps.



(a) *Mean TPOT (ShareGPT, 256)* **(b)** *P99 TPOT (ShareGPT, 256)* **(c)** *P99.9 TPOT (ShareGPT, 256)*

(d) *Mean TPOT (Zeta, 256)* **(e)** *P99 TPOT (Zeta, 256)* **(f)** *P99.9 TPOT (Zeta, 256)*

(g) *Mean TPOT (ShareGPT, 512)* **(h)** *P99 TPOT (ShareGPT, 512)* **(i)** *P99.9 TPOT (ShareGPT, 512)*

(j) *Mean TPOT (Zeta, 512)* **(k)** *P99 TPOT (Zeta, 512)* **(l)** *P99.9 TPOT (Zeta, 512)*

Figure 7. *Simulation performance on scaled clusters with 256 and 512 decode instances.*

Table 3. *Optimal assignment ratio comparison across different cluster scales.*

Dataset	# Inst.	RR	ILS	Sibyl
ShareGPT	256	56.6%	30.6%	**90.6%**
	512	61.7%	19.9%	**91.6%**
Zeta	256	29.1%	10.1%	**40.4%**
	512	50.0%	0.08%	**57.5%**

Sibyl achieves superior precision, maintaining > 90% accuracy on ShareGPT and significantly outperforming baselines on Zeta. Notably, ILS frequently underperforms the load-agnostic RR (*e.g.*, 10.1% vs. 29.1% on Zeta-256). This counter-intuitive result confirms that acting on *stale* information (ILS) creates severe "herding," where multiple requests are routed to the same seemingly idle instance, whereas Sibyl's predictive lookahead effectively resolves this contention.